

Bank Account Verification & Name Fetching APIs

Complete Guide on Free Public APIs, Penny Drop, Penniless & Provider Integration (India)

Prepared for MK-GOLD ERP & Billing System | September 2026

1. Overview: Free Public APIs vs. Restricted Bank Verification

A frequent requirement in gold-buying and billing ERPs is validating bank credentials entered by customers or branch staff. In India, banking APIs fall into two distinct regulatory tiers:

- **Tier 1: Free Public Banking APIs:** Open APIs (such as Razorpay IFSC) that require no API keys or banking licenses. They validate bank branches, city, district, state, MICR, and supported payment rails (NEFT, RTGS, IMPS, UPI). They cannot access account numbers or customer identities.
- **Tier 2: Regulated Account Validation (BAV):** Protected under Reserve Bank of India (RBI) and NPCI privacy laws. Accessing an account holder's legal name requires authorized Bank Account Validation (BAV) via licensed payment aggregators using Penny Drop, Penniless NPCI queries, or Reverse Penny Drop.

2. Verification Methods Comparison

- **Penny Drop (IMPS):** Sends ₹1.00 (or ₹0.01) to the customer's account via IMPS. Destination bank responds within 1-3 seconds with UTR and the registered beneficiary name (e.g., "BHARATH KUMAR").
- **Penny-Less Verification:** Direct inquiry via NPCI rails verifying account existence and name without moving money. Avoids micro-deposit accounting and is faster.
- **Reverse Penny Drop (UPI):** Merchant generates a dynamic UPI QR code. Customer pays ₹1 from their UPI app; NPCI webhook instantly returns the verified name and linked bank account.

3. Comparative Summary of Leading Indian Providers

Provider	Method	Speed	Name Match	Avg Cost / Check	Best For
RazorpayX	Penny Drop + Reverse	1-2 sec	Yes (0-100 Score)	Rs 1.50 - 2.50	Easiest setup & UI
Cashfree BAV	Penny Drop + Penniless	Sub-second	Yes (Fuzzy Match)	Rs 1.20 - 2.00	High volume & speed
Setu (Pine Labs)	Penny Drop + UPI	1-2 sec	Yes (AI match)	Rs 1.50 - 2.20	Enterprise KYC suite
Decentro	Penny Drop + Penniless	1-2 sec	Yes (Score)	Custom / Tiered	Full banking SDKs
Surepass.io	Penny Drop + Penniless	1-3 sec	Yes (Verified)	Rs 1.00 - 1.80	NBFCs & Gold loans

Detailed Commercial BAV API Specifications

Commercial Endpoints, Request/Response Payloads & Fuzzy Name Matching

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Provider 1: RazorpayX Bank Account Validation API

RazorpayX provides instant fund account validation. The system verifies if the account is active and fetches the exact bank-registered account holder name.

API Endpoint & Request:

```
POST https://api.razorpay.com/v1/fund_accounts/validations
Headers:
  Authorization: Basic base64(YOUR_KEY_ID:YOUR_KEY_SECRET)
  Content-Type: application/json

Payload:
{
  "account_number": "409000392345", // Your RazorpayX payout virtual account
  "fund_account": {
    "account_type": "bank_account",
    "bank_account": {
      "name": "Bharath Kumar",
      "account_number": "10710100283243",
      "ifsc": "CIUB0000107"
    }
  },
  "amount": 100, // 100 paise = INR 1.00
  "currency": "INR"
}
```

API Response:

```
{
  "id": "fav_J92Ksl01kd92La",
  "status": "completed",
  "validation_results": {
    "account_status": "active",
    "registered_name": "BHARATH KUMAR",
    "name_match_score": 100
  },
  "utr": "425118920192"
}
```

Provider 2: Cashfree Bank Account Verification (BAV)

Cashfree offers sub-second verification latency across all public and private Indian banks with built-in fuzzy name matching.

API Endpoint & Request:

```
POST https://api.cashfree.com/verification/bank-account/sync
```

```
Headers:
```

```
x-client-id: <YOUR_APP_ID>  
x-client-secret: <YOUR_SECRET_KEY>  
Content-Type: application/json
```

```
Payload:
```

```
{  
  "name": "Bharath Kumar",  
  "phone": "9876543210",  
  "bank_account": "10710100283243",  
  "ifsc": "CIUB0000107"  
}
```

API Response:

```
{  
  "status": "SUCCESS",  
  "subCode": "200",  
  "account_status": "VALID",  
  "account_exists": "YES",  
  "name_at_bank": "BHARATH KUMAR",  
  "name_match_score": "1.0",  
  "utr": "319201948201"  
}
```

Free Public Banking APIs & ERP Implementation

Free IFSC Lookup, Free Sandbox Testing & MK-GOLD Workflow

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1. Free Public Bank & IFSC Lookup API (Razorpay IFSC)

A completely free public API provided by Razorpay to validate IFSC codes, detect the bank and branch name, branch address, MICR code, and supported transfer modes without any API keys or account setup.

Free Public Endpoint:

```
GET https://ifsc.razorpay.com/{IFSC_CODE}

// Example Request:
GET https://ifsc.razorpay.com/CIUB0000107
```

Free API JSON Response:

```
{
  "BANK": "City Union Bank",
  "IFSC": "CIUB0000107",
  "BRANCH": "TAMBARAM",
  "ADDRESS": "NO. 15, RAJAJI ROAD, TAMBARAM, CHENNAI - 600045",
  "CITY": "CHENNAI",
  "DISTRICT": "KANCHEEPURAM",
  "STATE": "TAMIL NADU",
  "MICR": "600054012",
  "UPI": true,
  "IMPS": true,
  "NEFT": true,
  "RTGS": true,
  "BANKCODE": "CIUB"
}
```

2. Free Developer Sandbox & Testing Environments

- **Cashfree Free Sandbox:** <https://sandbox.cashfree.com/verification/bank-account/sync> — Allows unlimited free mock testing of account status and registered name responses without real money deduction.
- **Sandbox.co.in Free Tier:** Provides 100 free verification API calls every month for development, testing bank accounts, IFSC, and UPI VPA verification.
- **RazorpayX Test Mode:** Enables instant testing with simulated test account numbers to test success, failure, and name mismatch conditions at zero cost.

3. Operational Integration Workflow for MK-GOLD ERP

- **1. Instant IFSC Branch Validation (Free):** When branch staff types the IFSC code in the Customer Bank popup, the frontend calls <https://ifsc.razorpay.com/{IFSC}> to auto-fill Bank Name and Branch Name instantly at 0 cost.
- **2. Account Verification & Name Fetch:** On clicking "Verify Account", the server runs BAV to fetch the true registered beneficiary name (e.g., "BHARATH KUMAR").
- **3. Auto-Lock Account Holder Name:** The verified name is auto-filled into the "Account Holder Name" input and locked to read-only, preventing manual typos and fraud.
- **4. Pre-Payment Validation:** Ensures customer bank disbursements only go to active accounts under the customer's verified KYC identity.

